Introduction to Computational Methods for Digital Humanists

VU MA Digitale Geisteswissenschaften, University of Graz, Summer term 2027, Wednesdays, 10:15 – 11:45 A.M.

Session	Title	Content and Learning Objectives	Preparation for the session
Session 1 03.03.27	**Welcome & organization**	Organization: content, learning objectives, Moodle; the learning environment and your peers. Brainstorm research ideas: what data do you work with and what data do you want to work with? Let's look at text, image, audio, and video data Overview of the computational-methods pipeline in digital humanities: corpus, model, annotate, analyze, evaluate, reflect, disseminate.	Reading: Semester welcome letter Check: Minimal technical requirements for the class
Session 2 10.03.27	**Corpus Building I: Corpora in the computational humanities**	What is a corpus? What is a subcorpus? Which kinds of corpora exist? Where to find existing corpora for your computational humanities research area: text corpora and image and audio corpora, for example an IIIF image collections, GLAM and museum APIs, and oral-history archives Copyright and FAIR data; text-and-data-mining exceptions; derived text formats.	Reading: Bläß (2024[2020]), "Korpusbildung" (forTEXT journal; "Corpus Building" English translation on request). Schöch (2026), "Derived Text Formats" Instructional reading: How to work in groups Group work: find two to three partners and choose a group on Moodle and formulate your group-work agreement
Session 3 17.03.27	**Corpus Building II: sources, rights, acquisition**	Formulate research questions and hypotheses. Corpus design, file types, and documentation. Acquiring text: the Wikipedia API, Project Gutenberg, IIIF endpoint; what may and may not be redistributed (CC BY-SA, public domain, copyright).	Instructional reading: How to prepare the portfolio Read and try: Milligan, "Automated Downloading with Wget" (Programming Historian) + Jupyter notebook fetch_wikipedia_tei.ipynb Brainstorm: which corpus would you like to build? Group work: start building your corpus and define your research question
—	**Time off**	Public holiday / semester break (22.03.-03.04.2027)	
Session 4 07.04.27	**Corpus Building III: metadata and text preparation**	From raw text to structured text (known from TEI P5 basics in "X-Technologien I"). Text preparation pipeline: from HTML to TEI. Metadata enrichment: beyond the metadata from the HTML (Metadata standards such as Dublin Core or MODS, known from "Informationsmodellierung")	Reading: Drucker (2021), DH Coursebook, ch. 4 "Metadata, markup, and data description" Read and try: Momen et al. (2025) "Filling the temporal void" + Jupyter notebook metadata_enrichment.ipynb Group work: continue building your corpus
13.04.27	**Deadline: Portfolio Part 1**	Submit your first portfolio chapter (research question and corpus description) as a PDF on Moodle.	
Session 5 14.04.27	**Data Preparation: Data cleaning, tokenization, sentence splitting**	Cleaning and preparing text for computational analysis. Word tokenization, sentence segmentation, regular expressions, etc.	Reading: Jurafsky/Martin (2026), Speech and Language Processing, ch. 2 "Words and tokens" Group work: add metadata to your corpus
Session 6 21.04.27	**Annotation I: Object and named entity recognition**	What is annotation? Named entity recognition for text data, focus on entity types (PER, ORG, LOC, GPE). Beyond text data: visual analogue in object detection; Annotation as interpretation; theory pluralism and the guideline-dependence of categories. Intersubjectivity as the goal of annotation.	Reading: Ehrmann et al. (2023) "Named Entity Recognition and Classification in Historical Documents: A Survey" Group work: brainstorm whether named entity recognition is a useful method for your research question; experiment with the text cleaning and preparation steps (Jupyter notebooks)

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Session 7 28.04.27	**Annotation II: Annotation-based operationalization**	How to get from a humanities phenomenon to its operationalization for computational text analysis. Introduction to systematic manual annotation.	Reading: Pichler/Reiter (2022), "From Concepts to Texts and Back: Operationalization as a Core Activity of Digital Humanities" Group work: brainstorm about the phenomenon you want to analyze to answer your research question
Session 8 05.05.27	**Annotation III: Formulating intersubjective annotation guidelines, Evaluation**	Iterative preparation of annotation guidelines: Tagset creation and guideline formulation. What are compulsory and facultative parts of annotation guidelines? Inter-annotator agreement measurements	Reading: Reiter & Guhr (2025), "Instructions for Creating Annotation Guidelines" (optional) Reading: Andresen & Zinsmeister (2019), Korpuslinguistik (excerpt on "Corpus annotation" (English translation on request) Group work: brainstorm about useful annotation for approaching your research question
Session 9 12.05.27	**Annotation IV: Automated annotation with transformer models, evaluation**	Operationalization from phenomenon to automated annotation pipeline: training and test data preparation, transformer model family (text (BERT), vision (vision transformers) and speech (wav2vec, Whisper); Evaluating automatic annotation: precision, recall, F1, accuracy	Reading: Jurafski/Martin (2026), Speech and Language Processing, ch. 7 "Transformers and Pretraining" (optional) Reading: Vaswani et al. (2023[2017]), "Attention is all you need" Group work: formulate your annotation guidelines and prepare some manual annotations on your corpus data
Session 10 19.05.27	**Annotation V: Distant reading vs. distant viewing**	On the methodological and theoretical framework for the study of large collections of visual materials	Reading: Arnold/Tilton (2019), "Distant viewing: analyzing large visual corpora" Group work: brainstorm about multimodal approaches to your research question
Session 11 26.05.27	**Annotation VI: Large language model systems for annotation**	Prompting large language model systems for systematic annotation with schema-constrained output (json); The model's intrinsic schema vs. your guideline; responsible use and its limits. Running local open-weight models via LM Studio	Reading: Pagel/Pichler (2026), "Evaluating Humanities Theory Alignment in Large Language Models: Incremental Prompting and Statistical Assessment" Group work: finalize your annotation guidelines, prepare gold annotations for model fine-tuning or few-shot prompting
Session 12 02.06.27	**Data Analysis: Analysis and (multimodal) bias**	Analysis of your data with different approaches; large language model systems as a judge experiments; face analysis or speech recognition; Bias in data, research questions, and results; responsible AI; theory pluralism revisited.	Reading: Haltermann/Keith (2026), "Codebook LLMs: Evaluating LLMs as Measurement Tools for Political Science Concepts" Group work: play around with open-weight large language model systems and compare output to your gold annotations
08.06.27	**Deadline: Portfolio Part 2**	Submit your second portfolio chapter (operationalization and guidelines for manual annotation) as a PDF on Moodle.	
Session 13 09.06.27	**Poster preparation**	What is a scientific poster and how do I prepare it?	Group work: discuss what you see in your data analysis results, formulate your interpretation and prepare your material as short bullet points for poster preparation
Session 14 16.06.27	**Data interpretation: Answering the research question**	What do you see in your data analysis results? Accompanied writing session: Write up your data analysis and interpretation section with direct feedback	Group work: refine your group poster and prepare your presentation
20.06.27	**Deadline: Poster Submission**	Submit your poster by midnight as a PDF on Moodle.	
Session 15 23.06.27	**Conference simulation**	Group project presentations: poster slam and gallery walk; Feedback and open questions.	Instructional reading: How to ask and answer questions in a conference setting? Group work: prepare the presentation of your poster by equally distributed parts

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Session 16 30.06.27	**Conclusion and Outlook: Master – what's next?**	Summary of the class, reflection and peer feedback; Conclusion and outlook: what comes after your MA in Digital Humanities? Screening of recent job postings / real position announcements Overview in retrospect: what skills did you learn over the semester? Where do you fit? What could be your next step? How will computational methods support you in your future everyday work life?	Brainstorm: What did you learn over this semester? Look for job postings that could be interesting for you Group work: discuss and write down notes from the peer feedback;
15.07.27	**Deadline: Portfolio Part 3**	Submit your third portfolio chapter (reflection of your personal learning over the semester) as a PDF on Moodle.	